

Manish Kumar Lakhiwal

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PROFESSIONAL SUMMARY

Full Stack Developer and ML Engineer with experience across Java/Spring Boot, Python/Django, and Flutter. Currently building a Flutter + Django fintech platform (seller onboarding, invoice management, wallet). Published IEEE researcher on ML for network traffic classification and DDoS detection (97%+ accuracy). Comfortable moving across backend APIs, mobile UI, and ML pipelines.

TECHNICAL SKILLS

Languages: Python, Java, Dart, SQL, JavaScript (ES6+)

Backend / Web: Django, Django REST Framework, Spring Boot, Flask, REST APIs, JWT Auth

Mobile: Flutter, Dart, Material 3

ML / AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, Keras, Whisper, Pyannote, Hugging Face Transformers

Databases: PostgreSQL, MongoDB, MySQL, SQLite

Tools / DevOps: Git, Docker, Render, AWS, Cloudinary

Data / BI: Pandas, NumPy, Tableau, Power BI

PROFESSIONAL EXPERIENCE

Full Stack Developer — *Finworks360, New Delhi*

January 2026 – Present

- Developing a Flutter mobile app and Django backend for a fintech invoice discounting platform currently in build phase, with focus on the seller onboarding flow, invoice management, and wallet system
- Building REST APIs with Django REST Framework backed by PostgreSQL; handling transaction concurrency using Django's `select_for_update()` to prevent duplicate settlements
- Implemented Flutter UI with Material 3 dynamic theming, skeleton loaders, Hero transitions, and page route animations; optimized rendering using `RepaintBoundary` and `Impeller`
- Refactored legacy Django template architecture and consolidated CSS/JS; contributed to backend security hardening including OTP password reset and session management

Assistant Professor — *Shivalik College of Engineering, Dehradun*

July 2025 – December 2025

- Taught Machine Learning, Data Science, Python, and DBMS to 100+ undergraduates; designed labs on Scikit-learn, TensorFlow, feature engineering, and model evaluation; contributed to NBA accreditation documentation

Data Analyst — *Keptra Technologies (Remote)*

April 2025 – July 2025

- Built Python ETL pipelines (Pandas, NumPy) processing 100K+ firewall and network traffic records; developed anomaly detection scripts with real-time alerting and Tableau dashboards presenting security insights to stakeholders

Software Developer — *ScoreMe Solutions, Gurugram*

June 2021 – November 2021

- Built REST APIs in Spring Boot (Java) handling 10K+ daily requests for a credit-assessment platform, with JWT authentication and microservices architecture via Spring Cloud
- Engineered a Java-based ETL pipeline extracting structured financial data from 1000+ PDF bank statements daily, reducing a manual 4-hour process to roughly 30 minutes and enabling same-day credit decisioning

KEY PROJECTS

ai-meeting — **Local AI Meeting Transcription App** — *Flask, Whisper, Pyannote, DistilBART*

- Built a fully local (on-device) meeting transcription and summarization app using Whisper (ASR), Pyannote (speaker diarization), and DistilBART (summarization); added Hindi and Devanagari script support with custom Unicode handling

DDoS Attack Detection in SDN using ML — *Python, TensorFlow, Scikit-learn*

M.Tech Thesis

- Trained ensemble models (Random Forest, XGBoost, Deep Neural Networks) on 2.67M traffic records, achieving 97.2% accuracy and 0.93 F1-score classifying DDoS attacks; used RFE, SMOTE, and information-gain feature selection to reduce false positives
- Deployed on a Ryu SDN controller for real-time packet inspection with sub-100ms latency; results published in IEEE conference proceedings

EDUCATION

M.Tech in CSE (AI/ML) — *Manav Rachna International Institute Of Research & Studies*

2023 – 2025

B.Tech in CSE — *Maharshi Dayanand University, Rohtak*

2016 – 2020

PUBLICATIONS

- [Comparative Analysis of ML Techniques for Network Traffic Classification](#) — **IEEE ICCSC-2026**
- [Hybrid ML and Time Series Approaches for GHI Forecasting Using NSRDB Data](#) — **IEEE ICCSC-2026**
- [Forecasting PM2.5, NO2 in Gurugram Using Prophet](#) — CRC Press (Taylor & Francis)
- [Interface for Monitoring Pregnant Women's Fetal Health Based on Cardiotocogram](#) — Taylor & Francis